Introduction to Computational Thinking and Python Programming

Instructor: Sarom Leang, PhD (Teacher/Professor)

Specialist: Ravi Teja Talluri

Innovation Hall 203

Pronouns

- Instructor: Sarom Leang, Ph.D. (Teacher/Professor)
- Specialist: Ravi Teja Talluri

Schedule

- 10:00 AM – 10:15 AM (Homeroom)
- 10:15 AM – 11:35 AM (G1)
- 11:40 AM – 12:35 PM (Lunch)
- 12:40 PM – 02:00 PM (G2)

Student Expectations

- **NO FOOD**
- **NO DRINKS** (on the table)
- Be respectful to individuals and property
- Be open to learning
- Be open to not understanding
- Be patient with yourself
- Ask questions
- Explore
- **Embrace failure**

Course Overview (cont.)

Exploratory Topics

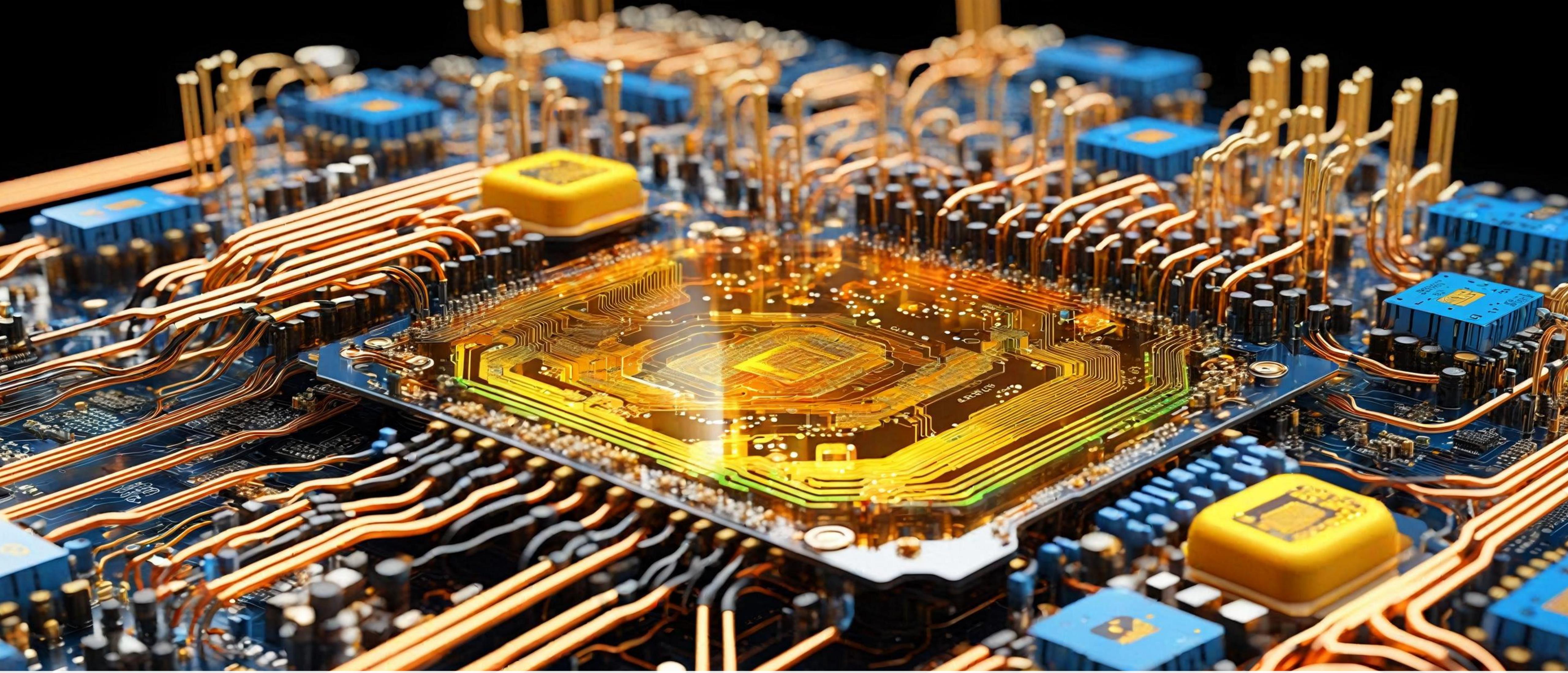
- ~~Session #1 – November 2, 2024~~
  - ~~Entrance Survey~~
  - ~~How Computers Work~~
- ~~Session #2 – December 7, 2024~~
  - ~~Parallel Computing~~
- ~~Session #3 – February 1, 2025~~
  - ~~Generative Artificial Intelligence~~
- Session #4 – March 22, 2025
  - Quantum Computing
- Session #5 – April 12, 2025
  - Future of Computing
  - Exit survey

Topics

- Algorithms and Problem Solving
- Debugging and Troubleshooting
- Loops
- Algorithms and Syntax
- Variables and Conditionals
- Variable Arithmetic
- Conditionals (If/Else)
- Compound Conditionals

Recap: Session 3

- Generative AI
  - A type of artificial intelligence that can create new content that is similar to the original data it was trained on
  - Learn patterns and relationships in data, and uses that knowledge to generate new content
  - Can produce content that is incorrect or incoherent (hallucinate)
  - Generate responses based on statistical patterns rather than comprehension
- Data (raw data) -> Information (processed) -> Knowledge (insight)
- Understand the rules governing generative AI use in your classes/schools/work place



Quantum Computing

Quantum Computing

Classical Computing

- Data is stored as bits (0 or 1)
- Data is processed **sequentially**
- Processing power increases **linearly** with the number of bits (32-bit processor, 64-bit processor,

Quantum Computing

- Data is stored as qubits (**superposition** of 0 and 1)
- Data can be processed in **parallel** (**superposition** and **entanglement**)
- Processing power increases **exponentially** with the number of qubits

Superposition

- Describes a state where a quantum system (e.g., qubits) **exists** in a **combination** of **multiple states simultaneously**
    - Classical bit: a coin that can be either heads **OR** tails
    - Qubit: a special coin that can be both heads **AND** tails at the same time

Entanglement

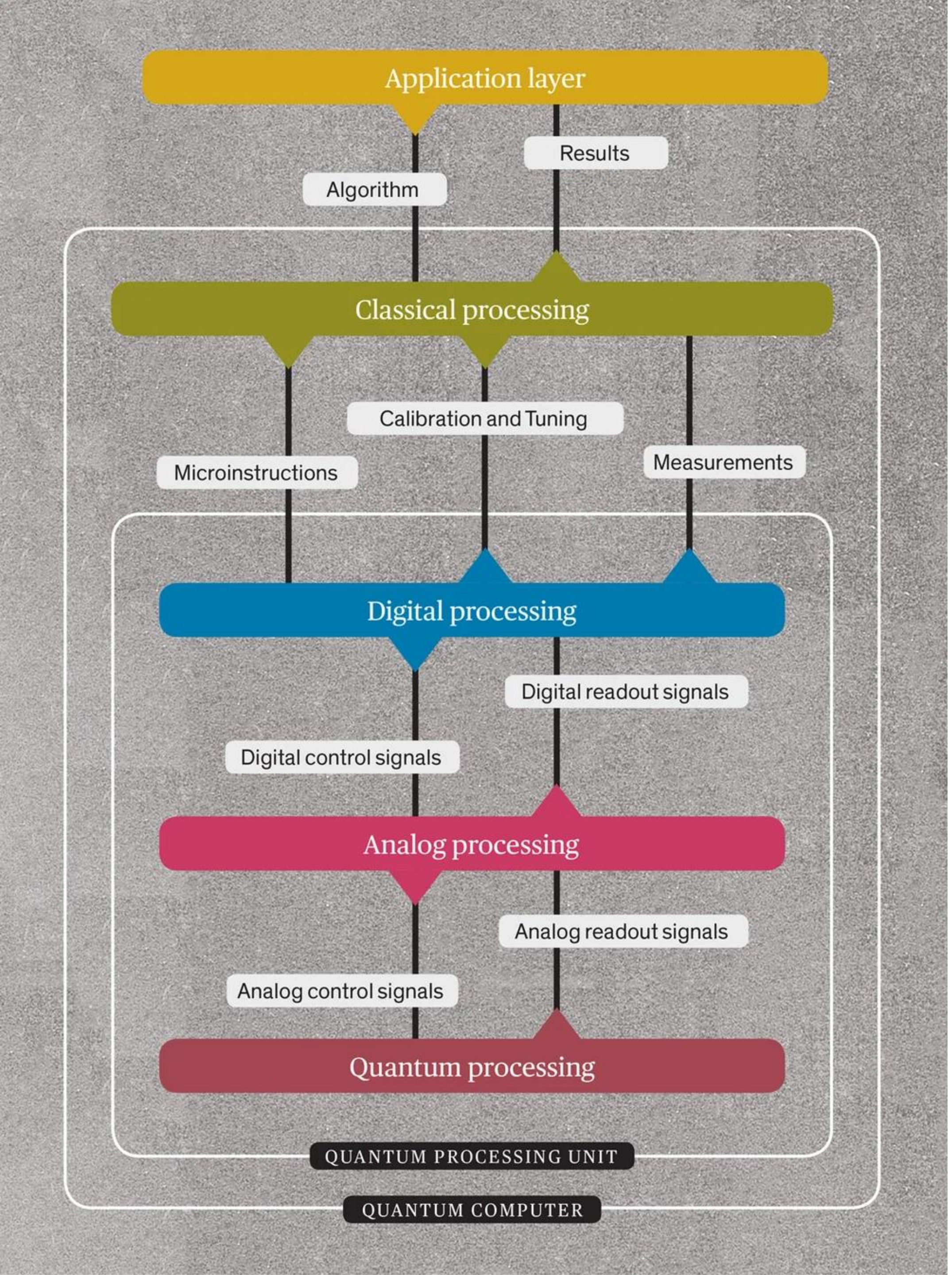
- Is a special **connection between two quantum particles** that lets them "talk" to each other **instantly**, no matter how far apart they are
  - Classical communication: sending a message from New York to Los Angeles via email or phone
  - Quantum entanglement: sending a message from New York to Los Angeles instantly, without using any physical medium

Superposition and Entanglement

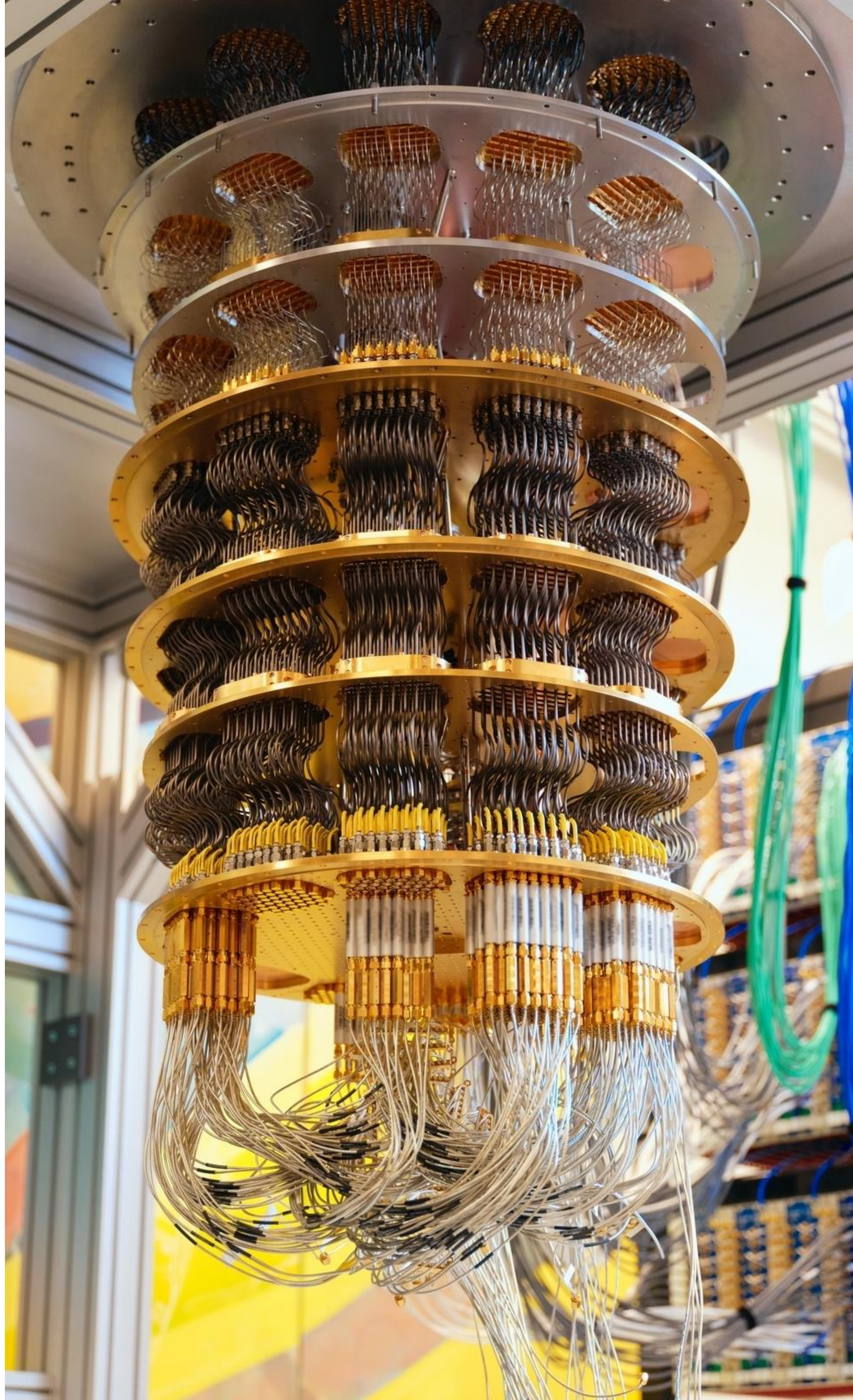
- Superposition allows for **processing multiple possibilities simultaneously**
  - Combination lock with 10 possible combinations
    - Classical – try each combination one at a time (sequential/serial)
    - Quantum – could try all 10 combinations simultaneously (parallel)
- Entanglement enables **quantum parallelism**
  - Two entangled qubits you can perform operations on both qubits simultaneously, effectively processing four possibilities at once: (00, 01, 10, and 11)
  - With three entangled qubits, eight possibilities at once: (000, 001, 010, 011, 100, 101, 110, 111)

Power of Quantum Computing

- Exponential scaling
  - As the number of qubits increases, the power of quantum computing grows exponentially – each qubit doubles the number of possible states that can be processed simultaneously
  - 1 qubit: 2 possible states (0 and 1)
  - 2 qubits: 4 possible states (00, 01, 10, and 11)
  - 3 qubits: 8 possible states (000, 001, 010, 011, 100, 101, 110, and 111)
  - n qubits: 2^n possible states
- Ideal applications
  - Large dataset – process massive amounts of data simultaneously (ML or DA)
  - Complex simulations – simulate complex systems
  - Cryptography – create unbreakable quantum encryption methods



Quantum Computer

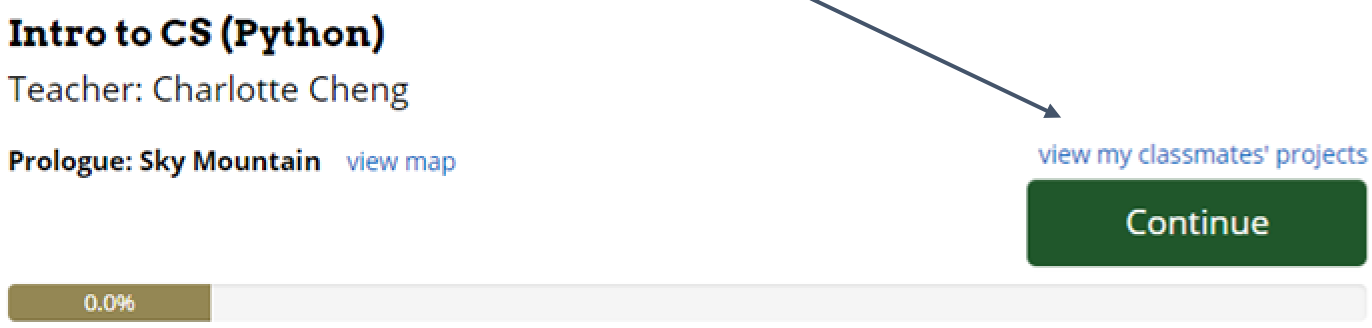


Today

- Log into Ozaria
- Complete Chapters 1-4

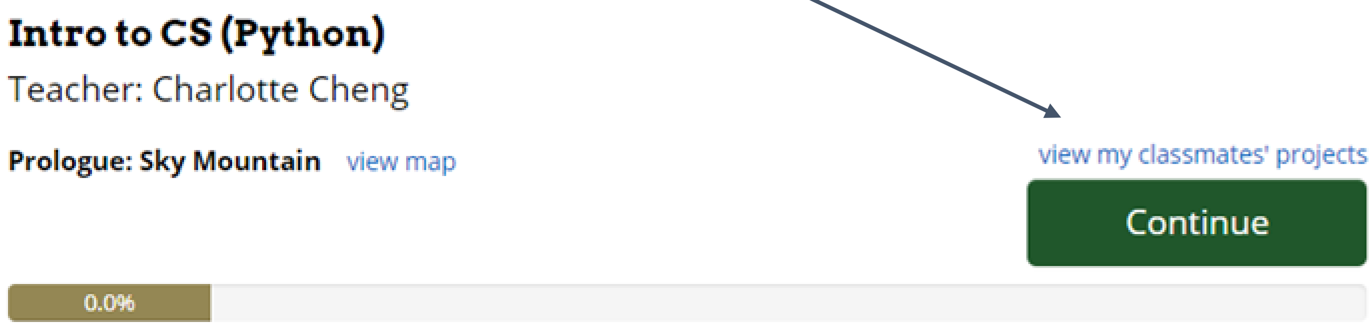
Logging into Ozaria (G2) - Python

1. Go to: https://www.ozaria.com
2. Click "**Sign Up**"
3. Enter in your **class code**
4. **Create an account** or login with your Google account
5. Make sure to **write down your user name**!
6. Click **Continue** to start playing



Logging into Ozaria (G2) - Javascript

1. Go to: https://www.ozaria.com
2. Click "**Sign Up**"
3. Enter in your **class code**
4. **Create an account** or login with your Google account
5. Make sure to **write down your user name**!
6. Click **Continue** to start playing



Innovation 203 Computer Issues